Tutorial - 2

1) A mobile Service provider wants to classify SMS messages as spam or not spam using a single layer neural network with 5 hidden neurons in the layer.

x_1 = Specific works & phrases

x_2 = links & attachments

x_3 = bias

x_1	x_2	x_3	Target
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

	w_1	w_2	b
h_1	0.1	-0.2	1
h_2	0.3	-0.1	1
h_3	0.2	0.4	1
h_4	0.1	0.3	1
h_5	-0.2	0.4	1

$v_1 = 0.1$

$v_2 = 0.2$

$v_3 = -0.3$

$v_4 = -0.1$

$v_5 = 0.4$

$$\boxed{z_1 = x_1 w_1 + x_2 w_2 + b}$$

$$z_1 = 4(0.1) + 2(0.2) + 1 = 1$$

$$z_2 = 4(0.3) + 2(-0.1) + 1 = 2$$

$$z_3 = 4(0.2) + 2(0.4) + 1 = 2.6$$

$$z_4 = 4(0.1) + 2(0.3) + 1 = 2$$

$$z_5 = 4(-0.2) + 2(0.4) + 1 = 1$$

Hidden neurons.

$$h_1 = \sigma(z_1) = \frac{1}{1+e^{-z}} = 0.731$$

$$h_2 = \sigma(z_2) = 0.880$$

$$h_3 = \sigma(z_3) = 0.930$$

$$h_4 = \sigma(z_4) = 0.880$$

$$h_5 = \sigma(z_5) = 0.731$$

$$z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5$$

$$= 0.73 \times 0.1 + 0.88 \times 0.2 + 0.93 \times (-0.3) + 0.88 \times (-0.1) + 0.73 \times 0.4$$

$$\boxed{z_{out} = 0.174}$$

$$\hat{y} = \sigma(z_{out}) = \frac{1}{1+e^{-0.174}} = 0.543$$

$\hat{y} \geq 0.5 \Rightarrow 1$ $\quad \hat{y} = 1 \quad y = 1$

Else $\Rightarrow 0$

$$\boxed{y = \hat{y}}$$

Sample (2) :- (1, 0, 1)

$z_1 = 1(0.1) + 0(-0.2) + 1$

$= 1.1$

$z_2 = 1(0.3) + 0(-0.1) + 1 = 1.3$

$= 1.3$

$z_3 = 1(0.2) + 0(0.4) + 1 = 1.2$

$z_4 = 1(0.1) + 0(0.3) + 1 = 1.1$

$z_5 = 1(-0.2) + 0(0.4) + 1 = 0.8$

Hidden neurons

$h_1 = 0.750$

$h_2 = 0.785$

$h_3 = 0.768$

$h_4 = 0.750$

$h_5 = 0.689$

$z_{out} = 0.750 \times 0.2 + 0.785 \times 0.2 + 0.768 \times (-0.3) + 0.750 \times (-0.1) + 0.689 \times 0.4$

$\boxed{z_{out} = 0.2022}$

$\hat{y} = 0.550$ $\hat{y} = 1$ $y = 0$

$\boxed{y \neq \hat{y}}$

<u>updating weights</u>

$y = 0$ $\hat{y} = 0.55$ $\eta = 0.1$

$\eta(y - \hat{y}) = 0.1(0 - 0.55) = 0.55$

$$\boxed{v_i^{new} = v_i^{old} + \eta(y - \hat{y}) h_i}$$

$v_1^{new} = 0.1 + (-0.055)(0.750) = 0.05875$

$v_2^{new} = 0.2 + (-0.055)(0.785) = 0.1568$

$v_3^{new} = -0.3 + (-0.055)(0.768) = -0.34224$

$v_{ij}^{new} = -0.1 + (-0.055)(0.760) = -0.14125$

$v_0^{new} = 0.4 + (0.055)(0.689) = 0.362105$

Sample (3) :- (6,3,1)

$z_1 = 6(0.1) + 3(-0.2) + 1 = 1$

$z_2 = 6(0.3) + 3(-0.1) + 1 = 2.5$

$z_3 = 6(0.2) + 3(0.4) + 1 = 3.4$

$z_4 = 6(0.1) + 3(0.3) + 1 = 2.5$

$z_5 = 6(-0.2) + 3(0.4) + 1 = 1$

hidden neurons

$h_1 = 0.731$

$h_2 = 0.924$

$h_3 = 0.967$

$h_4 = 0.924$

$h_5 = 0.731$

$z_{out} = 0.731(0.1) + 0.924(0.2) + 0.967(-0.3) + 0.924(-0.1) + 0.731(0.4)$

$\boxed{z_{out} = 0.1678}$

$\hat{y} = 0.541 \quad \hat{y} = 1 \quad y = 1$

$\boxed{y = \hat{y}}$

$z_1 = 2(0.1) + 1(-0.2) + 1 = 1$

$z_2 = 2(0.3) + 1(-0.1) + 1 = 1.5$

$z_3 = 2(0.2) + 1(0.4) + 1 = 1.8$

$z_4 = 2(0.5) + 1(0.3) + 1 = 1.5$

$z_5 = 2(-0.2) + 1(0.4) + 1 = 1$

hidden neurons

$h_1 = 0.731$

$h_2 = 0.817$

$h_3 = 0.858$

$h_4 = 0.817$

$h_5 = 0.731$

$z_{out} = 0.731(0.1) + 0.817(0.2) + 0.858(-0.3) + 0.817(-0.1) + 0.731(0.4)$

$\boxed{z_{out} = 0.1898}$

$\hat{y} = 0.547 \Rightarrow \hat{y} = 1 \quad y = 0 \quad \boxed{y = \hat{y}}$

updating weights

μ $\pm 0.04.2e$

$y = 0 \quad \hat{y} = 0.547 \quad \eta = 0.1$

$\eta(y - \hat{y}) = 1.1(0 - 0.547) \Rightarrow -0.0547$

$v_1 = 0.1 + (-0.0547)(0.731) = 0.060$

$v_2 = 0.2 + (-0.0547)(0.817) = 0.155$

$v_3 = -0.3 + (-0.0547)(0.858) = -0.346$

$v_4 = -0.1 + (-0.0547)(0.817) = -0.144$

$v_5 = 0.4 + (-0.0547)(0.731) = 0.360$